

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS1

SUBJECT: Geography

TOPIC: The Earth & Solar System

Introduction

- Geography is the study of the earth — its physical features, people and the relationships between them. It is divided into physical geography (landforms, climate, vegetation) and human geography (population, settlement, economic activities).

The solar system

- The solar system has eight planets — Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune — orbiting the sun, along with moons, asteroids and comets. The earth is the third planet from the sun, at an average distance of about 150 million km (1 astronomical unit).

The earth's structure

- The earth is a slightly flattened sphere. It has three zones: the crust (thin outer layer of rocks), the mantle (hot, semi-molten) and the core (inner core solid, outer core liquid). The lithosphere (crust + upper mantle) floats on the asthenosphere.

Importance of geography

- It explains the distribution of resources, weather and population; it builds map-reading skills and an understanding of environmental problems like desertification, flooding and erosion.

CLASS EVALUATION

1. List the eight planets in order from the sun.
2. Name the three zones of the earth's interior.
3. State two branches of geography.
4. What is the astronomical unit?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS1

SUBJECT: Geography

TOPIC: Latitude & Longitude

Introduction

- Latitude and longitude form the grid system used to fix the position of places on the earth's surface. They are measured in degrees (°) and minutes (').

Latitude

- Latitude is the angular distance of a place north or south of the equator (0°). Important lines: the equator (0°), Tropic of Cancer (23½°N), Tropic of Capricorn (23½°S), Arctic Circle (66½°N), Antarctic Circle (66½°S) and the poles (90°N/S). Latitudes are also called parallels.

Longitude

- Longitude is the angular distance east or west of the Prime Meridian (0°), which passes through Greenwich, London. The 180° meridian is the International Date Line. Places east of Greenwich see the sun earlier; the world is divided into 24 time zones, each 15° of longitude = 1 hour.

Calculating time and distance

- Time: every 15° = 1 hour (add for east, subtract for west). Distance: 1° of latitude ≈ 111 km. Example: Lagos (3°E) and London (0°) differ by 3° = 12 minutes; when it is 12:00 in London it is 12:12 in Lagos.

CLASS EVALUATION

1. Define latitude and longitude.
2. Calculate the time difference between two places 30° of longitude apart.
3. Name the five important lines of latitude.
4. Why are latitudes are called parallels?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS1

SUBJECT: Geography

TOPIC: Earth Movements

Introduction

- The earth rotates on its axis once every 24 hours and revolves round the sun once every 365¼ days. These two movements produce most of the rhythms of life.

Rotation (daily)

- Rotation is the spinning of the earth from west to east, causing day and night, the apparent rising and setting of the sun, deflection of winds (Coriolis effect) and tidal movements. Places east of the Greenwich meridian experience sunrise first.

Revolution (annual)

- Revolution is the earth's journey round the sun along its elliptical orbit, with its axis tilted at $23\frac{1}{2}^{\circ}$. It causes the changing seasons, the unequal lengths of day and night, and the migration of the overhead sun between the Tropics of Cancer and Capricorn.

Equinoxes and solstices

- **Equinoxes** — 21 March and 23 September: the sun is overhead the equator; day and night are equal everywhere. **Solstices** — 21 June (sun overhead the Tropic of Cancer, longest day in the north) and 22 December (sun overhead the Tropic of Capricorn).

CLASS EVALUATION

1. Distinguish between rotation and revolution.
2. What causes the seasons?
3. Explain an equinox.
4. State three effects of the earth's rotation.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS1

SUBJECT: Geography

TOPIC: Rocks & Minerals

Introduction

- A rock is a natural solid mass of minerals; a mineral is a naturally occurring inorganic substance with a definite composition and structure. The study of rocks is petrology.

The three rock classes

- **Igneous** — formed by cooling and solidification of magma/lava (granite, basalt).
- **Sedimentary** — formed from weathered particles or remains of organisms deposited in layers (sandstone, limestone, coal). Fossils are found here. They may be mechanical, organic or chemical.
- **Metamorphic** — formed when existing rocks are changed by heat and/or pressure (marble from limestone, gneiss from granite, slate from shale).

The rock cycle

- Rocks are continually transformed: weathering breaks igneous and metamorphic rocks into sediments; sediments compact into sedimentary rocks; heat and pressure change them to metamorphic; melting returns them to magma. There is no beginning or end to the cycle.

Uses of rocks and minerals

- Granite — building; limestone — cement, agriculture; coal — fuel; marble — sculpture, buildings; salt — food; gold, tin, columbite — jewellery, industry and foreign exchange.

CLASS EVALUATION

1. Name the three classes of rocks with two examples each.
2. How is marble formed?
3. Why are fossils found mainly in sedimentary rocks?
4. State two uses of limestone.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS1

SUBJECT: Geography

TOPIC: Weathering & Mass Movement

Introduction

- Weathering is the breaking down of rocks in place by physical, chemical and biological agents, without removal. It is the first stage of denudation.

Types of weathering

- **Physical/mechanical** — temperature changes (expansion and contraction), frost action, pressure release and abrasion; rocks crack into fragments.
- **Chemical** — oxidation, hydration, carbonation (rainwater + CO₂ dissolves limestone), solution and hydrolysis; the rock's minerals are changed.
- **Biological** — plant roots widening cracks, burrowing animals, and humic acids from decay.

Mass movement

- Mass movement is the downslope movement of weathered material under gravity: **slow** — soil creep, solifluction; **fast** — landslides, mudflows, rockfalls. It is aided by water saturation, steep slopes, earthquakes and the removal of vegetation.

CLASS EVALUATION

1. Distinguish between weathering and erosion.
2. Explain how carbonation weathers limestone.
3. What factors accelerate mass movement?
4. Give two agents of biological weathering.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS1

SUBJECT: Geography

TOPIC: Denudation & The Work of Rivers

Introduction

- Denudation is the wearing away of the land (weathering, mass movement, erosion, transportation and deposition). Running water — rivers — is the most important agent of erosion on land.

Work of a river

- Rivers erode by hydraulic action, corrosion/abrasion, solution and attrition; they transport material as solution, suspension, saltation and traction; they deposit when the load exceeds the capacity (slower water).

Landforms along the course

- **Youthful (upper) course** — V-shaped valleys, interlocking spurs, waterfalls, rapids, potholes.
- **Mature (middle) course** — meanders, river cliffs and slip-off slopes, ox-bow lakes, braided channels.
- **Old (lower) course** — levees, flood plains, deltas, estuaries.

CLASS EVALUATION

1. Define denudation and name its processes.
2. List four ways a river erodes.
3. Name two landforms of the youthful course.
4. How is an ox-bow lake formed?